SKYWAVE™
AN ORBCOMM COMPANY




# ST 91xx
# Hardware Guide

T413, Version
Mar 2026

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# PREFACE

## Purpose

This guide is as an overview of the hardware characteristics and specifications for the ST 9100 and ST 9101 products. The information in this guide refers to both products, hereinafter referred to as the "ST 91xx" unless noted otherwise.

## Notation

Hardware components and hardware labels in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Bulleted lists highlight information where order or sequence is not crucial.

## Reference

The content of the following documents might be useful in conjunction with this guide.

Document names and numbers are subject to change, or be discontinued, without notice.

| | | | |
|---|---|---|---|
| [T404] | LSF Developer Guide | [N210] | IsatData Pro Gateway Web Service 2 User Guide |
| [T405] | LSF API Reference Guide | [N206] | MTWS Cellular Protocol |
| [T414] | ST 91xx Installation Guide | [N214] | OGx Gateway Web Service (OGWS) User Guide |

## Battery Safety Warnings

CAUTION: DO NOT short circuit or expose the battery to temperatures above the maximum rated temperature.

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: Store in a cool, well-ventilated area. Elevated temperatures will result in shortened battery life.

CAUTION: DO NOT replace the battery. Changing the battery without ORBCOMM's permission could violate regulatory conformity.

CAUTION: DO NOT throw the internal battery or the device into fire.


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# 1 PRODUCT OVERVIEW

The ST 91xx is a flexible, robust, and programmable dual mode satellite-cellular terminal. It is ideal for remotely monitoring and controlling fixed and portable assets in industries as diverse as transportation, oil and gas, utilities, maritime and more. The versatile, environmentally sealed ST 91xx is ideal for rugged environments in the world's most remote areas.

The ST 91xx (Figure 1) is a satellite-cellular terminal. Features include the following:

- A satellite-cellular transceiver for communicating with the network
  - ST 9100:
    - Part number ST9100-D01 for use in the Americas
    - Part number ST9100-C01 for use outside of the Americas
    - Part number ST9100-E01 for Saudi Arabia
  - ST 9101:
    - Part number ST9101-F01
- An integral multi-GNSS subsystem
- Four (4) general purpose I/Os
- Two (2) dedicated open drain outputs
- Four (4) inputs (digital or 0-5 V analog of which the first two can be configured as 4-20 mA)
- Two (2) RS-232 ports
- One (1) RS-485 port
- Two (2) CAN bus ports
- One (1) 1-Wire interface
- 3-Axis 16-bit accelerometer
- Multiple SIM support
- Cellular module
- Bluetooth connectivity
- Internal backup battery
- Satellite antenna: p/n ST101599-APA (OGx) or ST901065-APA (IsatData Pro) standard antenna, ST101600-APA (OGx) or ST901066-APA (IsatData Pro) low elevation antenna
  - (optional) Satellite antenna with magnetic base p/n ST901065-APM (OGx) or p/n ST101065-APM (IsatData Pro)
- Cellular antenna:
  - ST 9100: p/n ST101066-001
  - ST 9101: p/n ST101561-001
- Terminal shroud (optional - p/n ST101014-001)


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Figure 1: ST 91xx Satellite-Cellular Transceiver




The transceiver's built-in programmability allows it to work as a standalone data-messaging transceiver, with built-in I/O data collection and processing capabilities. Feature-rich software tools make programming easy and shorten the design and testing time. The transceiver can also be configured with terminal apps. Terminal apps are configurable device-level applications that include specific feature sets that are implemented by SKYWAVE. Contact Partner Support or your Account Manager for further details.

## 1.1 Overview of the Messaging System

The satellite messaging system is designed to support the management of mobile or fixed assets located around the world. An asset fitted with one of SKYWAVE's satellite based mobile terminals can have their status and locations monitored and send large messages.

The network provides the following key features and benefits:

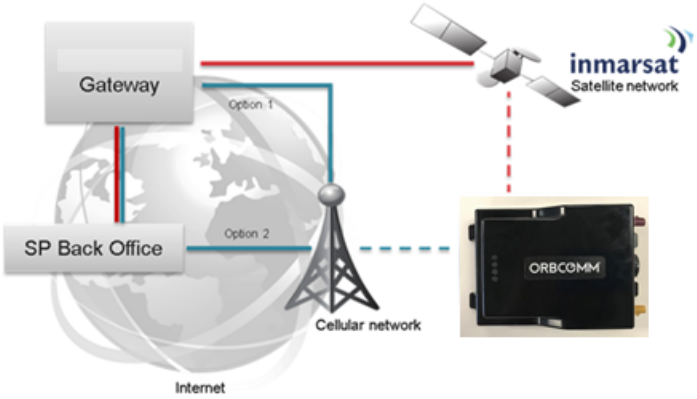
- Polling of terminal status and location
- Scheduled reporting of terminal status and location
- Transmission of text messages to and from a serial port on the transceiver
- Two-way communication for messaging to and from the asset for near real-time control
- Up to 6,399 bytes from-mobile messages
- Up to 10,000 bytes to-mobile messages
- Default acknowledged messages
- Global service

Service is provided to end users by Solution Providers (SPs) who use the satellite network to offer particular applications and/or services to their clients. The SPs link their application services to the satellite terminals by connecting to the gateway. This acts as the communications hub of the system, routing traffic to and from the terminals and the various service providers.

The terminal can be configured to route cellular messages through the same gateway that supplies satellite messages. This is shown as Option 1 in the System Diagram. Option 2 represents a terminal configured to route cellular messages directly to a customer or Solution Provider proprietary cell server. In this case the connection to SKYWAVE's OGx or IsatData Pro gateway supplies satellite messages. Refer to [T404, N206, N210 or N214] for more information about configuring the terminal's cellular messaging transport.


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Figure 2: System Architecture




The satellite-cellular terminal is based on Lua software and is supported by a suite of network tools, enabling SPs a programmable platform they can tailor to their specific applications.

## 1.2 Terminal

Note: Hardware components may not be exactly as shown in this document. A terminal consists of a transceiver unit plus antennas.

Transceivers with a standard antenna operate on either the OGx or IsatData Pro network[1] at an elevation angle of 20° to 90° and -5° to 90° for transceivers with low elevation antennas. The transceivers are self-contained, compact, and provide low power consumption.

A cellular module is available to operate over the cellular network

The transceiver's built-in programmability allows it to work as a stand-alone with built-in I/O data collection and processing capabilities. Terminals are suitable for the AVL market.

Feature-rich software tools make application design easy and shorten the design and testing time. SKYWAVE also provides consulting services to SPs to help program the transceiver and get customer applications running quickly.

## 1.3 Transceiver Components

CAUTION: Do not rely solely on the terminal for emergency (SOS) calls.

In addition to the features mentioned earlier, the transceiver has the following benefits:

- Designed to be used as a standalone or incorporated into an SP solution
- Built-in multi-GNSS receiver to calculate position, speed, and heading
- Quick and easy installation reduces labor time and costs
- Installed firmware
- Flexible custom application design (Lua Services Framework)
- Wide operational temperature range
- Satellite plus cellular modem integration

[1] Network depends on antenna type and network registration.


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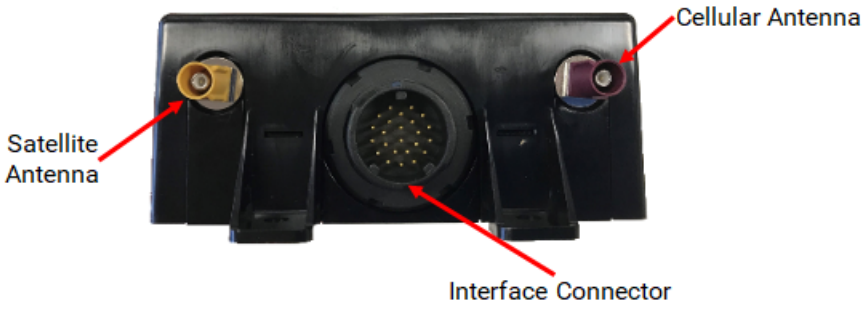
- Discrete I/O ports to interface with a wide range of after-market accessories
- Rugged construction

### 1.3.1 Transceiver Unit

Each transceiver is a self-contained unit, including a satellite / cellular modem, a multi-GNSS module, programmable microcontroller, and multiple discrete and analog I/Os (input / output) capable of monitoring and controlling external sensors and devices. Ideal for mobile applications, it is also suitable for fixed installations.

Arranging the transceiver unit's connectors (Figure 3) at one end of the unit simplifies installation. Sturdy flanges on the side make mounting quick and easy.

Figure 3: Connector Position



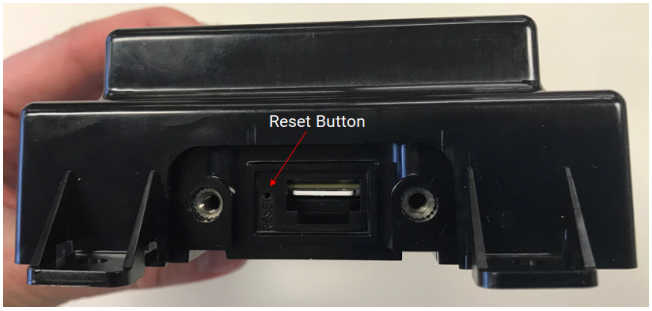

An anti-tamper SIM door on the back side of the transceiver (Figure 4) provides easy access to the SIM card holder and reset button.

Figure 4: SIM Access Door




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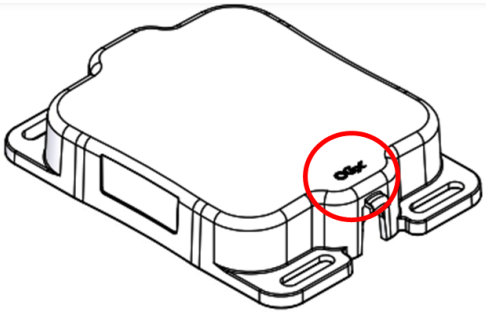
Figure 5: Reset Button




### 1.3.2 Satellite Antenna

CAUTION: Use only approved color-matching antennas.

Note: The OGx satellite antennas have an identifying mark near the connector.



The satellite-cellular transceiver's satellite antenna is waterproof and designed to operate in extreme environments. It has four mounting flanges for installation.

The satellite antenna connects to the transceiver using a 5 m (16 ft.) cable terminated with a curry yellow colored FAKRA RF connector.

The satellite-cellular transceiver is available with either the standard satellite antenna (Figure 6) or the low elevation satellite antenna (Figure 7).


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Figure 6: Standard Satellite Antenna



Figure 7: Low Elevation Satellite Antenna



### 1.3.3 Cellular Antenna

CAUTION: Use only color-matching antennas.

The bar cellular antenna is an LTE antenna with a burgundy-colored FAKRA connector.

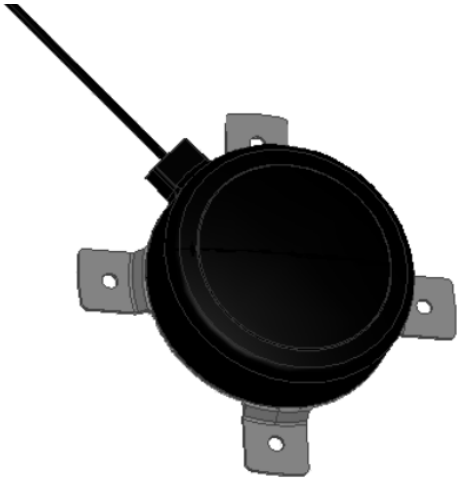

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Figure 8: Bar Cellular Antenna




The puck cellular antenna with a burgundy-colored FAKRA connector.

Figure 9: Puck Cellular Antenna




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### 1.3.4 Terminal Shroud

Use the optional terminal shroud if mounting the transceiver outdoors.

Figure 10: Terminal Shroud




### 1.3.5 ST 91xx Cables and Connectors

The following are available for the ST 91xx:

- A 5-meter blunt cut cable (p/n ST101062-002). Refer to [T414] for details.
- An IP67 Field Installable Connector (p/n ST101096). Refer to [T414] for details.
- A development cable (p/n ST101084-001). Refer to APPENDIX A for details.

CAUTION: An external 5 A slow blow fuse must be added in series with the external voltage wire (Table 2).


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# 2 SPECIFICATIONS

## 2.1 Temperature

| Parameter | Value |
|---|---|
| Operating Temperature Range | -40°C to +85°C (-40°F to +185°F) |
| Recommended Storage Temperature Range | -20°C to +35°C (-4°F to +95°F) |

### 2.1.1 Internal Backup Battery Temperature

Table 1 defines the internal backup battery's temperature specifications.

Table 1: Transceiver with Internal Backup Battery Temperature Specifications

| Parameter | Value |
|---|---|
| Charge Temperature Range | 0°C to +45°C (32°F to +113°F) |
| Discharge Temperature Range | -20°C to +75°C (-4°F to +167°F) |
| Storage Range | |
| ≤ 1 month | -20°C to +45°C (-4°F to +113°F) |
| ≤ 3 months | -20°C to +35°C (-4°F to +95°F) |
| ≤ 1 year | 0°C to +30°C (32°F to +86°F) |
| Ideal for long-term storage | 10°C to +25°C (60±25% R.H.), (50°F to +77°F) |

## 2.2 Electrical

### 2.2.1 Input Range

CAUTION: An external 5 A slow blow fuse must be added in series with the external voltage wire (Table 2).

| Parameter | Value |
|---|---|
| Power Supply Voltage | 9 to 32 V DC |

### 2.2.2 Power Consumption

Following are power consumption values for 12 VDC and 24 VDC.

#### 2.2.2.1 Power Consumption - 12 VDC

Typical values with a transceiver input voltage of 12 VDC.

ST 9100

| Operation Mode | Other Conditions | 25°C (77°F) | -40°C (-40°F) | 85°C (185°F) |
|---|---|---|---|---|
| Sleep | Externally powered | 212 µA | 203 µA | 500.98 µA |
| Shell | Console connected | 73.71 mA | 101.32 mA | 99.38 mA |


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| Operation Mode | Other Conditions | 25°C (77°F) | -40°C (-40°F) | 85°C (185°F) |
|---|---|---|---|---|
| SatCom Tx | Burst current | 733.87 mA | 720.77 mA | 728.13 mA |
| SatCom Rx | Burst current for Rx freq. 1540045000 1000 2 (C/No=42dBHz) | 76.70 mA | 25.77 mA | 80.13 mA |
| GPS | Cold fix current during u-blox on command | 32.60 mA | 38.37 mA | 36.59 mA |
| Cell Tx | A typical cell Tx event | The average current consumption for a Tx event is 0.8 mA when the device is powered with 8 V.<br>The power consumption stays linear across the device's 9 to 32 V operating voltage range and -40°C to +85°C (-40°F to +85°F) temperature span. | | |

ST 9101

| Operation Mode | Other Conditions | 25°C (77°F) | -40°C (-40°F) | 85°C (185°F) |
|---|---|---|---|---|
| Sleep<br>(Externally Powered) | Externally powered, battery connected, and charger disabled | 228.67 µA | 317.28 µA | 409.03 µA |
| SatCom Tx | Tx burst current<br>Event generated using Messenger's to-mobile and from-mobile messages.<br>Reports without GPS (SIN 19, MIN 1) | 0.98 A | 1 A | 0.98 A |
| SatCom Rx | Rx burst current<br>Event generated using Messenger's to-mobile and from-mobile messages.<br>Reports without GPS (SIN 19, MIN 1) | 121 mA | 123 mA | 121 mA |
| GPS | GPS Current<br>Event generated using Lua script for requesting position using the console. | 76 mA | 81 mA | 76 mA |
| Cell Tx | A typical Cell Tx Event | The average current consumption for a Tx event is 0.8 mA when the device is powered with 8 V.<br>The power consumption stays approximately linear across the device's 9 to 32 V operating voltage range and -40°C to +85°C (-40°F to +85°F) temperature span. | | |

### 2.2.2.2 Power Consumption - 24 VDC

Typical values with a transceiver input voltage of 24 VDC.


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ST 9100

| Operation Mode | Other Conditions | 25°C (77°F) | -40°C (-40°F) | 85°C (185°F) |
|---|---|---|---|---|
| Sleep | Externally powered | 221 µA | 217 µA | 369.92 µA |
| SatCom Tx | Burst current | 353.486mA | 356.46 mA | 356.43 mA |
| SatCom Rx | Burst current for Rx freq. 1540045000 1000 2 (C/No=42dBHz) | 47.28 mA | 45.45 mA | 49.58 mA |
| GPS | Cold fix current during u-blox on command | 21.74 mA | 24.91 mA | 23.12 mA |
| Cell Tx | A typical cell Tx event | The average current consumption for a Tx event is 0.8 mA when the device is powered with 8 V.<br>The power consumption stays linear across the device's 9 to 32 V operating voltage range and -40°C to +85°C (-40°F to +85°F) temperature span. | | |

ST 9101

| Operation Mode | Other Conditions | 25°C (77°F) | -40°C (-40°F) | 85°C (185°F) |
|---|---|---|---|---|
| SatCom Tx | Tx burst current<br>Event generated using Messenger's to-mobile and from-mobile messages.<br>Reports without GPS (SIN 19, MIN 1) | 470 mA | 473 mA | 467 mA |
| SatCom Rx | Rx burst current<br>Event generated using Messenger's to-mobile and from-mobile messages.<br>Reports without GPS (SIN 19, MIN 1) | 69 mA | 72 mA | 70 mA |
| GPS | GPS Current<br>Event generated using Lua script for requesting position using the console. | 46 mA | 47 mA | 47 mA |
| Cell Tx | A typical Cell Tx Event | The average current consumption for a Tx event is 0.8 mA when the device is powered with 8 V.<br>The power consumption stays approximately linear across the device's 9 to 32 V operating voltage range and -40°C to +85°C (-40°F to +85°F) temperature span. | | |

### 2.2.2.3 Consumption with Battery at 8.2 V

ST 9100

| Operation Mode | Current Ambient | Current -20°C (-4°F) | Current 70°C (167°F) |
|---|---|---|---|
| Sleep | 250 µA | 237 µA | 575 µA |
| Shipping | 40 µA | 40 µA | 44 µA |


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ST 9101

| Operation Mode | Other Conditions | 25°C (77°F) | -20°C (-4°F) | 70°C (167°F) |
|---|---|---|---|---|
| Sleep | Battery-powered, external power disconnected | 236.00 µA | 318.45 µA | 444.00 µA |
| Shipping | | 59.93 µA | 55.28 µA | 64.46 µA |

### 2.2.3 Load Dump Protection

Active load dump protection is provided on the power pins. The cut-off is >34 V and automatic reset of the load dump occurs when the input voltage is <34 V.

### 2.2.4 Inrush Currents

Typical inrush currents: 12 volts and 25°C (77°F).

| Parameter | ST 9100 Value | ST 9101 Value |
|---|---|---|
| Peak in-rush current | 4.12 A | 3.87 A |
| In-rush pulse duration | 138 µS | 8.94 µS |

### 2.2.5 Reverse Voltage Input

| Parameter | Voltage |
|---|---|
| Reverse Polarity Protection | -40 V DC (maximum) |

### 2.2.6 External Supply Detection

The detector trigger voltage is dependent on the current drawn by the terminal.

| External Supply Voltage Falling (V) | Trigger Voltage (V) | Load Current (A) |
|---|---|---|
| 6.1 | 0.383 | 0.4 (Charging) |
| 6.4 | 0.386 | 0.8 (Satcom Tx) |
| 6.5 | 0.377 | 2.4 (Satcom+ Cell Tx) |

### 2.2.7 SIM Cards

The transceiver offers two embedded (not field replaceable) and one removable 3FF (micro-SIM) SIM card. The specifications for the SIM cards are the same.

| Parameter | Value |
|---|---|
| SIM Voltage | 1.8 V or 3 V standard SIM cards |
| Card Detection | Switch connected to cellular module |


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## 2.3 Connectors

| | |
|---|---|
| **Transceiver 24 position mating connector** | Chogori Technology Company |
| **Satellite Antenna** | IMS Connector Systems<br>3400.SMBA.2K10.089 (RG58/LMR-195 sized cable)<br>FAKRA - K-curry yellow |
| **Cellular Antenna** | IMS Connector Systems<br>3400.SMBA.2D10.029 9RG174 sized cable)<br>FAKRA - D-bordeaux |

### 2.3.1 Connector Pin Assignment

Table 2 maps to the layout shown in Figure 11.

Figure 11: Transceiver View of Connector

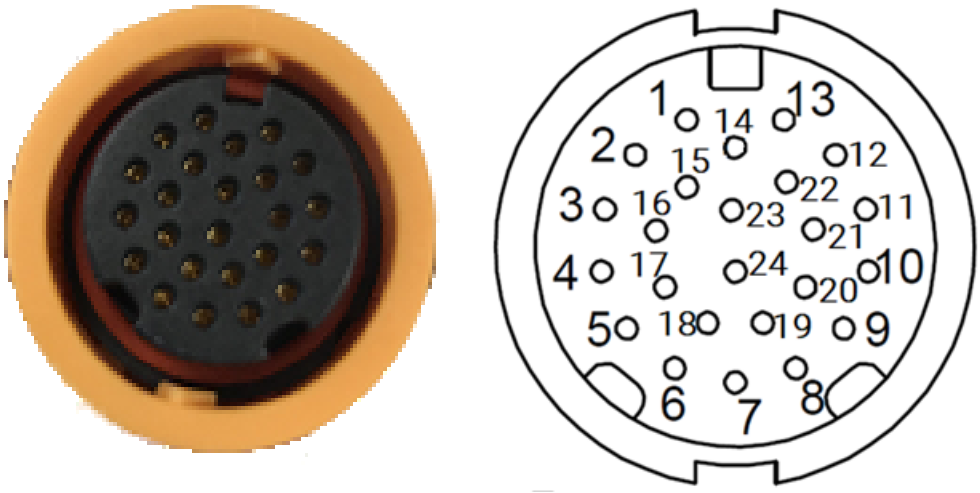



Table 2: Electrical Pin Assignment

| PIN | Function | Type | Description |
|---|---|---|---|
| 1 | RS485_A | I/O | Half duplex RS-485 driver output or receiver input (complementary to RS485_B) |
| 2 | Digital_IN4 / 0 to 5 V_IN4 | I | Digital input or 0 to 5 V analog input |
| 3 | Digital_IN3 / 0 to 5 V_IN3 | I | Digital input or 0 to 5 V analog input |
| 4 | I/O_4 | I/O | Multifunction GPIO, push-pull, analog input, current limited current sink or ignition load |
| 5 | I/O_2 | I/O | Multifunction GPIO, push-pull, analog input or current sink |
| 6 | Ground | PWR | External supply ground return |
| 7 | External Voltage | PWR | External 9 to 32 VDC supply |
| 8 | Output_6 | O | Open drain output |


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| PIN | Function | Type | Description |
|---|---|---|---|
| 9 | 1Wire Com | PWR | 1-Wire return path |
| 10 | Console_RS232_TX | O | Transmit port signals: ±15 kV ESD protected, RS-232 levels (nominally ±5.5 V) protected to ±25 V. Receiver inputs. |
| 11 | AUX_RS232_RX | I | Receive port signals: ±15 kV ESD protected, RS-232 levels (nominally ±5.5 V). Transmitter outputs. |
| 12 | CAN1_H | I/O | High level CAN bus line |
| 13 | CAN1_L | I/O | Low level CAN bus line |
| 14 | CAN0_L | I/O | Low level CAN bus line |
| 15 | RS485_B | I/O | Half duplex RS-485 driver output or receiver input (complementary to RS485_A) |
| 16 | Digital/Analog_IN1 /<br>0 to 5 V_IN1 /<br>P1_4 to 20 mA+ | I | Digital input or 0 to 5 V analog input or 4 to 20 mA input |
| 17 | I/O_3 | I/O | Multifunction GPIO, push-pull, analog input or current sink |
| 18 | I/O_1 | I/O | Multifunction GPIO, push-pull, analog input or current sink |
| 19 | Output_5 | O | Open drain output |
| 20 | 1Wire_DATA | I/O | Input / output driver for 1-Wire line |
| 21 | Console_RS232_RX | I | Receive port signals: ±15 kV ESD protected, RS-232 levels (nominally ±5.5 V). Transmitter outputs. |
| 22 | AUX_RS232_TX | O | Transmit port signals: ±15 kV ESD protected, RS-232 levels (nominally ±5.5 V) protected to ±25 V. Receiver inputs. |
| 23 | CAN0_H | I/O | High level CAN bus line |
| 24 | Digital_IN2 /<br>0 to 5 V_IN2 /<br>P2_4 to 20 mA+ | I | Digital input or 0 to 5 V analog input or 4 to 20 mA input |

## 2.4 I/O Interface

### 2.4.1 Standard General Purpose I/Os

The transceiver supports four (4) configurable general purpose I/Os (GPIO I/O_1 to I/O_4):

- Digital input with weak (1 MΩ) pull-down
- Digital input with 20 to 50 K pull-down
- Digital input with 20 to 50 K pull-up
- Analog input
- Digital output – push-pull
- Digital output – open drain
- Disabled

On certain vehicles, I/O_1 can be used to monitor ignition. The voltage on the I/O pin may not drop low enough to present a logic zero to the host processor when the ignition is turned off. In such a case I/O_1 can be configured to switch in a 4 kOhm load to draw the voltage below the logic zero threshold level of the host processor ensuring a logic zero. The other I/O ports should not be used for ignition monitoring. Refer to section 2.4.1.1


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I/O_4 provides dedicated overcurrent / short circuit protection circuitry when operated in the open drain mode. The other I/Os do not have this circuitry therefore I/O_4 is recommended for applications requiring overcurrent / short circuit protection.

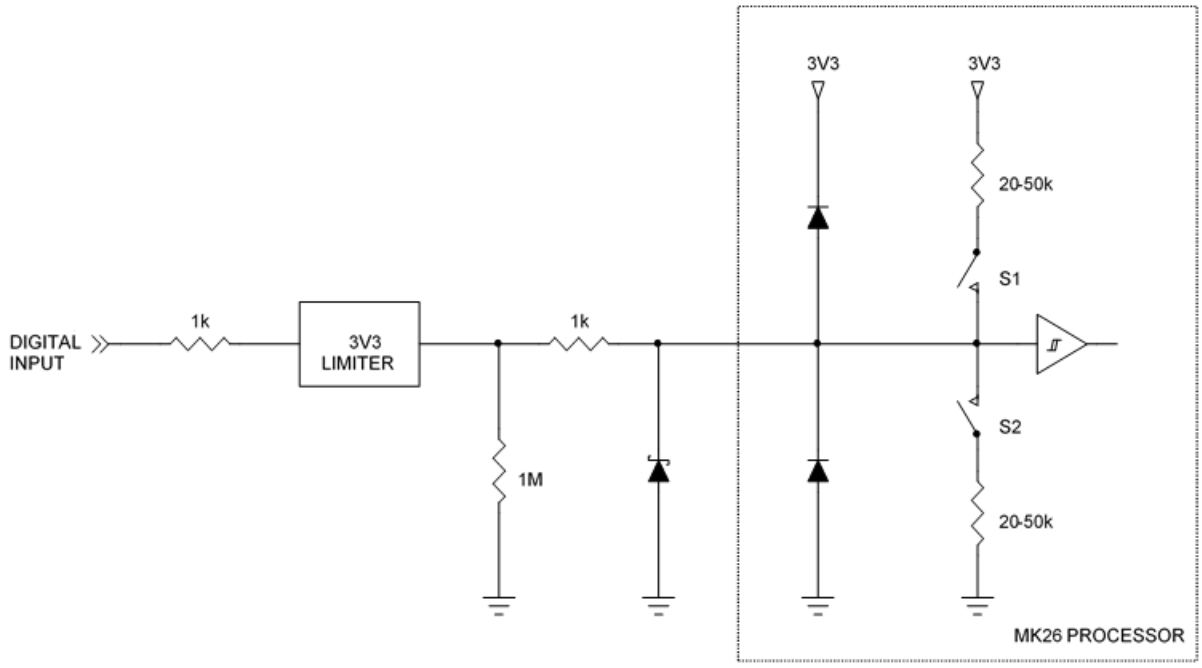
Simplified block diagrams of the I/O when configured as digital inputs, digital outputs, and analog inputs are shown in the figures below (Figure 12, Figure 13, and Figure 14).

The transceiver also supports two dedicated outputs (Output_5 and Output_6). More information on these outputs can be found in section 2.3.1.

#### 2.4.1.1 Digital Input

Figure 12 shows a schematic of the I/O when configured as a digital input.

Figure 12: Digital Input




| Input Type | S1 | S2 |
|---|---|---|
| With weak pull-down | Open | Open |
| With pull-down | Open | Closed |
| With pull-up | Closed | Open |

The input specifications are provided in the table below.

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Input low range | -10 | - | 1.16 | V |
| Input high range | 2.31 | - | 150 | V |

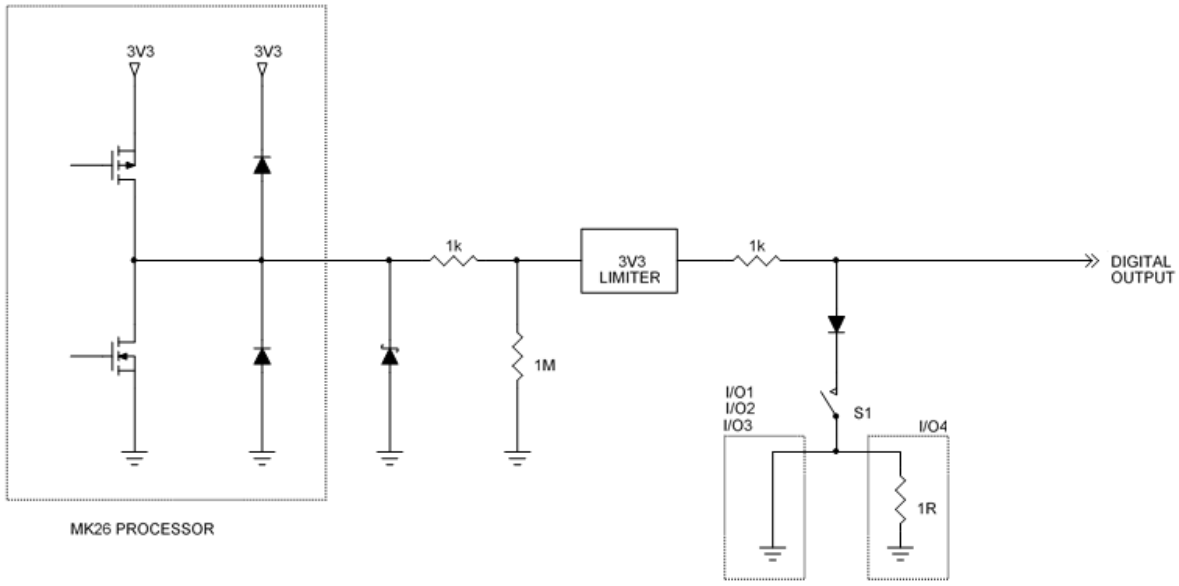

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| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Input current with weak pull-down (weak 1 MΩ pull-down still in place); $V_{in}$ = 3.3 V | - | 4.5 | - | µA |
| Input source current with 50 k pull-up ($V_{in}$ = 0.0 V) | - | 75 | - | µA |
| Input sink current with 50 k pull-down ($V_{in}$ = 3.3 to 150 V) | - | 81 | - | µA |
| Input bandwidth | 1 | - | - | kHz |

### 2.4.1.2 Digital Output

Figure 13 shows a schematic of the I/O when configured as a digital output.

Figure 13: Digital Output




| | |
|---|---|
| Push-pull | S1 = Open |
| Open drain | S1 = Closed (Low Impedance)<br>S1 = Open (High Impedance) |

#### 2.4.1.2.1 Push-pull

In the push-pull configuration the output is driven directly from the microprocessor.

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Output high voltage - open circuit | 3.23 | 3.3 | 3.37 | V |
| Output high voltage (sourcing 10 mA) | 2.8 | - | - | V |
| Output low voltage (sinking 10 mA) | - | - | 0.5 | V |
| Output bandwidth | 100 | - | - | Hz |


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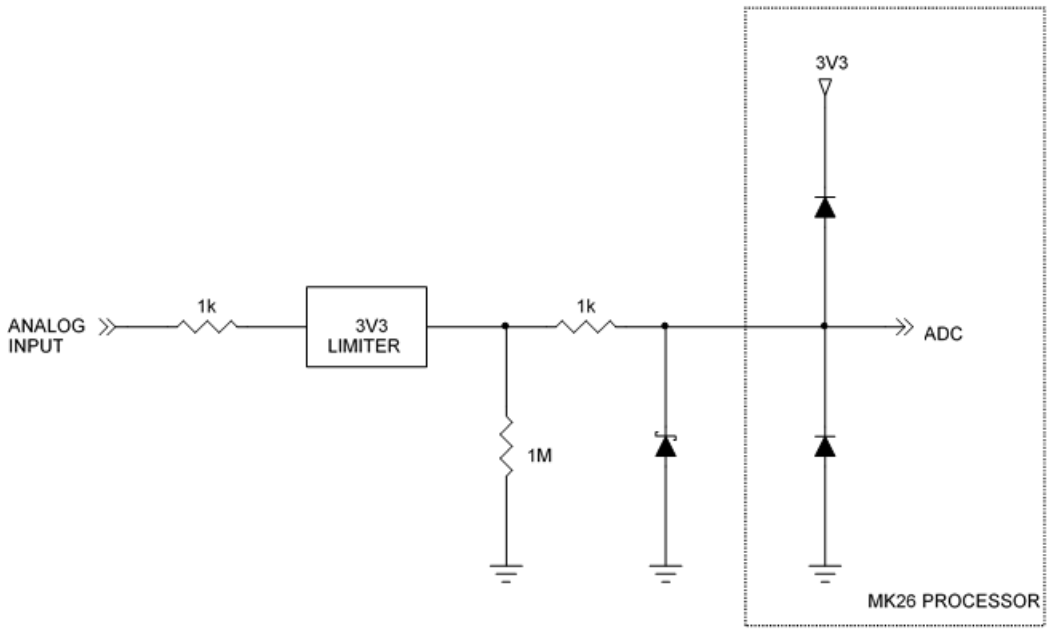
#### 2.4.1.2.2 Open Drain

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Sink current (do not exceed) | - | - | 250 | mA |
| Output voltage (sinking 250 mA)<br>I/O_1 to I/O_3<br>I/O_4 | <br>-<br>- | <br>1.67<br>1.92 | <br>2.00<br>2.25 | <br>V<br>V |
| Absolute limits (high impedance) | -10 | - | 150 | V |
| Output bandwidth | 100 | - | - | Hz |

### 2.4.1.3 Analog Input

Figure 14 shows a schematic of the I/O when configured as an analog input.

Figure 14: Analog Input



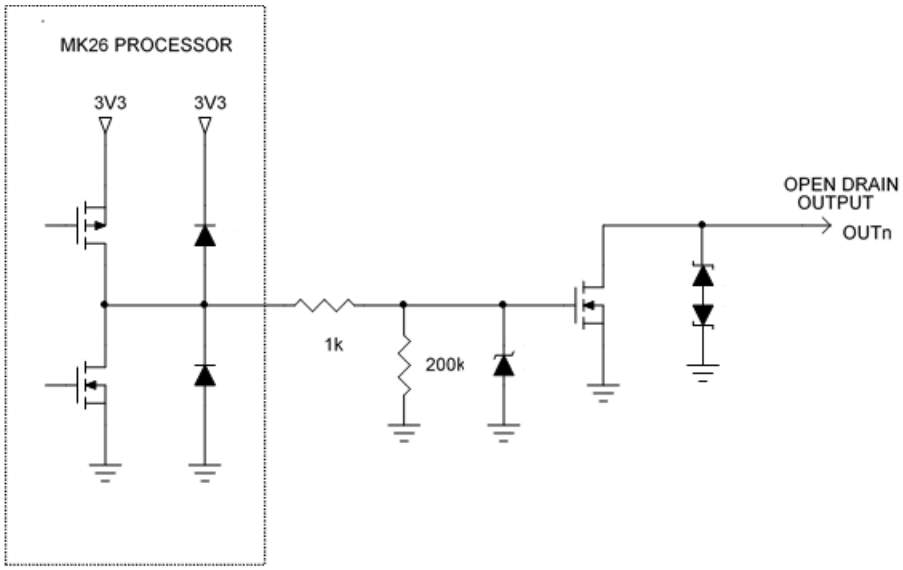

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Input impedance | - | 1 | - | MΩ |
| Normal input measurement range | 0 | - | 3.3 | V |
| Resolution (12 bits) | - | 0.8 | - | mV |
| Proportional measurement error | - | - | 3 | % |
| INL error | - | - | 2 | LSB |
| Absolute limits | -10 | - | 150 | V |


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## 2.4.2 Dedicated Outputs

The transceiver provides two open drain outputs (output_5 and output_6) that can be used to turn on various devices such as relays, lights or audible alarms. These outputs are capable of sinking current only. Both outputs are controlled from the host processor. The outputs are not protected against over current conditions, and you must ensure that the maximum current capability of the internal switch is not exceeded. Both outputs include ESD protection. You must also ensure that the voltage applied to the output pin does not exceed the maximum value as shown in the table below.

Figure 15: Open Drain Outputs




| Parameter | Max. | Units |
|---|---|---|
| Sink Current | 250 | mA |
| Applied Voltage | 40 | V |
| Internal switch power dissipation | 691 | mW |
| Voltage output (sink current = 250 mA) | 2.76<br>(minimum 48 mV) | V |

## 2.4.3 Multi-purpose Ports

In addition to the standard I/Os, the transceiver provides the following multi-function ports.

- Four (4) digital input only ports.
- Four (4) 0 to 5 V analog input only ports.
- Two (2) 4 to 20 mA inputs.

Four pins on the interface connector are independently configured to provide the following combinations:

- Four (4) digital inputs or,
- Four (4) 0 to 5 V analog inputs or,
- Two (2) 4 to 20 mA inputs or,
- Two (2) digital inputs and two 0 to 5 V analog inputs or,

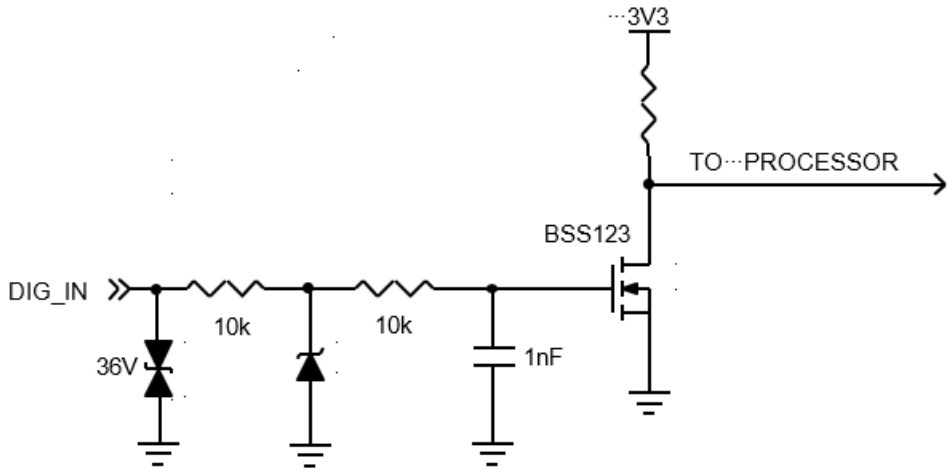

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- Two (2) digital inputs and one 4 to 20 mA input or,
- Two (2) 0 to 5 V analog inputs and one 4 to 20mA input.

### 2.4.3.1 Input Only Ports

Four ports (PINs 2, 3, 16, and 24) can be configured as dedicated inputs. Each input is ESD protected by a 36 V transient voltage suppressor that clamps the input transient at 58 V. A 15 V Zener ensures the FET maximum gate voltage of 20 V is not exceeded.

Figure 16: Dedicated Inputs




| Parameter | Min. | Max. | Processor | Units |
|---|---|---|---|---|
| **Digital Input** | | | | |
| Typical Input high voltage (Zener starts conducting at ±14.49 V) | 1.6 | 14 | 0 | V |
| Maximum input high voltage | - | 32 | 0 | V |
| Input low voltage | 0 | 1.4 | 3.3 | V |
| Input frequency | 1 | 10 | - | Hz |
| **ESD** | | | | |
| TVS breakdown voltage | 40 | 44.2 | - | V |
| TVS clamp voltage | - | 58.1 | - | V |

### 2.4.3.2 Analog Inputs (0 to 5 V)

Four ports (PINs 2, 3, 16, and 24) can be configured as dedicated 0 to 5 V analog inputs. 0 to 5V applied to the ports is converted to 0 to 3.3 V to be compatible with the ADC voltage range of the host processor.

| Parameter | Min. | Typ. | Max. | Units |
|---|---|---|---|---|
| **Analog Input** | | | | |
| Operating nput voltage range | 0 | - | 5 | V |


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| Parameter | Min. | Typ. | Max. | Units |
|---|---|---|---|---|
| Maximum withstand input voltage | - | - | 32 | V |
| Accuracy | -2 | 1 | 2 | % |
| ADC resolution(ENOB) | 12.2 | 13.9 | - | bit |
| Input resistance | - | 34243 | - | Ohm |
| Input current at 5 V | - | 146 | - | µA |
| **ESD** | | | | |
| TVS breakdown voltage | 40 | | 44.2 | V |
| TVS clamp voltage | - | | 58.1 | V |

### 2.4.3.3 Inputs (4 to 20 mA)

The ST 91xx can monitor two 4 to 20 mA sensors. Two ports (PINs 16 and 24) can be configured as two dedicated 4 to 20 mA receivers.

| Parameter | Min. | Typ. | Max. | Units |
|---|---|---|---|---|
| **Current Loop** | | | | |
| Operating current range | 4 | - | 20 | mA |
| Load voltage at 4 mA | 0.396 | - | 0.404 | V |
| Load voltage at 20 mA | 1.98 | - | 2.02 | V |
| Load resistance | 99 | - | 101 | Ω |
| Loop voltage (supplied by users externally) | 10 | - | 32 | V |
| Maximum input high voltage | - | - | 32 | - |
| Accuracy | -2 | 1 | 2 | % |
| ADC resolution (ENOB) | 12.2 | 13.9 | - | bit |
| **ESD** | | | | |
| TVS breakdown voltage | 40 | | 44.2 | V |
| TVS clamp voltage | - | | 58.1 | V |

## 2.5 Serial Interfaces

Transceivers have the following interfaces:

- 2 x CAN bus
- 1 x RS-485/J1708
- 2 x RS-232
- 1 x 1-Wire

### 2.5.1 CAN Bus

The transceiver provides two CAN bus interfaces for sending and receiving frames.

The transceiver incorporates a controller area network interface with signaling rates up to 1 Mbps.

Note: You must provide a termination resistor externally to the transceiver.


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For ST 9100:

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Input Common Mode Voltage | -7 | - | 12 | V |
| Differential Input Voltage | -6 | - | 6 | V |
| Peak to Peak Output Common Mode Voltage | - | 1 | - | V |
| Differential Output Voltage (dominant) | 1.2 | - | 3 | V |
| Differential Output Voltage (recessive) No Load | -0.5 | - | 0.05 | V |
| CANH or CANL | -36 | - | 36 | V |
| **ESD Protection (external to driver)** | | | | |
| Human Body Model / Air Gap | - | ±15 | - | kV |
| Contact Discharge (IEC 61000-4-2) | - | ±8 | - | kV |

For ST 9101:

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Input Common Mode Voltage | -30 | - | 30 | V |
| Differential Input Voltage | -30 | - | 30 | V |
| Peak to Peak Output Common Mode Voltage | - | 1 | - | V |
| Differential Output Voltage (dominant) | 1.5 | - | 3 | V |
| Differential Output Voltage (recessive) No Load | -0.05 | - | 0.005 | V |
| CANH or CANL | -70 | - | 70 | V |
| **ESD Protection (external to driver)** | | | | |
| Human Body Model / Air Gap | - | ±15 | - | kV |
| Contact Discharge (IEC 61000-4-2) | - | ±8 | - | kV |

### 2.5.2 RS-485/J1708

The transceiver provides a half-duplex RS-485 or J1708 interface as an accessory bus and for SCADA interfacing with signaling rates up to 250 kbps.

Note: You must provide a termination resistor externally to the transceiver when required.

The electrical characteristics of the interface are:

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Input Common Mode Voltage | -8 | - | 12.5 | V |
| Differential Input Threshold | -200 | - | 200 | mV |
| Output Common Mode Voltage | - | 1.8 | 3 | V |
| Differential Drive Output, 54 Ω load | 1.5 | 2.3 | - | V |
| **ESD Protection (external to driver)** | | | | |
| Human Body Model / Air Gap | - | ±15 | - | kV |
| Contact Discharge (IEC 61000-4-2) | - | ±8 | - | kV |


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### 2.5.3 RS-232 (Console and Auxiliary)

The two RS-232 interfaces default to the following settings: 9600 bit/s, 1 start, 8 data, 1 stop bit, and no parity. The baud rate is configurable up to 115,200 bps.

The electrical characteristics of the interface are:

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| Rx Input Low Threshold for DTE Connected | - | - | -2.7 | V |
| Rx Input High Threshold for DTE Connected | 2.7 | - | - | V |
| Rx Threshold for DTE Disconnected | -0.3 | - | 0.3 | V |
| Serial Rx Input Low Threshold | 0.6 | - | - | V |
| Serial Rx Input High Threshold | - | - | 2.4 | V |
| Rx Input Voltage Range | -25 | - | 25 | V |
| Serial Tx Low Output (3 kΩ load) | - | - | -3.7 | V |
| Serial Tx High Output (3 kΩ load) | 3.7 | - | - | V |
| **ESD Protection** | | | | |
| Human Body Model | - | ±15 | - | kV |
| Contact / Air Gap Discharge | - | ±8 | - | kV |

### 2.5.4 1-Wire

The 1-Wire interface allows connection to downstream 1-Wire devices connected on a bus, or to a single button reader. Relative to any attached 1-Wire device, the transceiver behaves as the master. The 1-Wire driver supports 3 or 5 V devices on the bus.

At standard speed, the 1-Wire supports up to 39 devices over a 61-meter (200 feet) CAT5 cable. In overdrive, the usable expected distance is reduced to ≤15 meters (≤50 feet) with a maximum node count of 9.

The electrical characteristics of the interface are:

| Parameter | Min. | Typical | Max. | Units |
|---|---|---|---|---|
| 1-Wire Input High Voltage | 1.8 | - | - | V |
| 1-Wire Input Low Voltage | - | - | 0.6 | V |
| 1-Wire Output Low Voltage | - | - | 0.2 | V |
| 1-Wire Load Current @3V | - | 4 | - | mA |
| **ESD Protection** | | | | |
| Human Body Model | - | ±15 | - | kV |

## 2.6 Satellite RF Specifications

| Parameter | OGx | IsatData Pro |
|---|---|---|
| Receive | | |
| Frequency Band | 1525 to 1559 MHz | 1525 to 1559 MHz |
| Modulation | OQPSK | OQPSK |


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| Parameter | OGx | IsatData Pro |
|---|---|---|
| Symbol Rate | 2000, 8000, 16000 symbols / second | 3000 symbols / second |
| Polarization | RHCP | RHCP |
| Transmit | | |
| Frequency Band | 1626.5 to 1660.5 MHz | 1626.5 to 1660.5 MHz |
| Modulation | OQPSK | OQPSK |
| Symbol Rate | 800, 1600, 3200, 6400, 12800 symbols / second | 900 symbols / second (maximum) |
| Polarization | RHCP | RHCP |

### 2.6.1 Satellite (Standard) Antenna

| Parameter | Value |
|---|---|
| Maximum EIRP | 7 dBW |
| Elevation Angle | 20° to 90° degrees |
| Maximum transmit antenna gain | 4.5 dBic |

### 2.6.2 Satellite (Low Elevation) Antenna

| Parameter | Value |
|---|---|
| Maximum EIRP | 5 dBW |
| Elevation Angle | -5° to 90° degrees |
| Maximum transmit antenna gain | 2.5 dBic |

### 2.6.3 Cellular Antenna

#### 2.6.3.1 LTE/3G/2G Bar Antenna (ST101066-001)

| Parameter | Value |
|---|---|
| Frequency Band | 700 to 960 / 1710 to 2170 / 2600 to 2700 MHz |
| Impedance | 50 Ω |
| Radiation Pattern | Omni |
| Max. Gain | 2.5 dBi (tested with 10 cm /4" cable) |

#### 2.6.3.2 LTE/3G/2G Puck Cellular Antenna (ST101561-001 and ST101561-002)

| Parameter | Value |
|---|---|
| Frequency | 617 to 2690 MHz |
| Impedance | 50 Ω |
| Polarization | Linear |
| Max. Gain | 3 dBi for 3-meter cable and 0.7 dBi for 5-meter cable |


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#### 2.6.3.3 LTE Puck Cellular Antenna (ST101335-001)[1]

| Parameter | Value |
|---|---|
| Frequency Band | 698 to 960 / 1710 to 2690 / 3300 to 3800 MHz |
| Impedance | 50 Ω |
| Polarization | Linear |

## 2.7 Satellite Transmitting Power

The maximum transmitting power (EIRP) for the IsatData Pro satellite is 7 dBW.

## 2.8 GNSS Module

The transceiver allows concurrent reception of up to three (3) GNSS channels.

The manufacturer's specifications are given in the table that follows.

Table 3: Multi-GNSS Specifications

| Parameter | GPS | GLONASS | BeiDou | Galileo |
|---|---|---|---|---|
| **Time to First Fix** | | | | |
| Cold Start | 29 s | 30 s | 34 s | 45 s |
| Warn Start | 2s | 2 s | 3 s | 7 s |
| Hot Start | 1 s | 1 s | 1 s | 1 s |
| **Sensitivity** | | | | |
| Tracking | -162 dBm | -166 dBm | -160 dBm | -159 dBm |
| Hot Start | -157 dBm | -156 dBm | -155 dBm | -151 dBm |
| Cold Start | -148 dBm | -145 dBm | -143 dBm | -138 dBm |
| **Accuracy** | | | | |
| Horizontal Position | 2.5 m | 4.0 m | 3.0 m | 4.0 m |
| Velocity | 0.05 m/s | | | |
| Heading | 0.3 degrees | | | |

## 2.9 Internal Backup Battery

The internal backup battery provides autonomous battery charging to the transceiver and operates directly from the external supply over the 9 to 32 VDC input range. The internal backup battery contains a protection card to ensure that the pack does not get damaged due to a short circuit, over discharge or an over-charge condition.

If the battery voltage is below the minimum set voltage, the charger turns off.

[1] Manufacturer datasheet 5-meter data.


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Table 4: Internal Backup Battery

| Parameter | Value |
|---|---|
| Battery Chemistry | Lithium Ion |
| Back-up Period | 48 hours with a reporting interval of 1 minute over cellular and 15 minutes over satellite |
| Rated Capacity | 2000 mAh |
| Charge (capacity) Retention | 90% (after 28 days at 25 ±5°C (77 ±9°F)) |
| Battery Cut-off | 7 V |
| Nominal Pack Voltage | 7.2 V (2 x cells in series) |
| Minimum Discharge Voltage | 5 V |
| Charging Voltage | 8.4 V |
| Peak Output Current | 6 A |
| Battery Weight | ~100 g (~3.5 oz) |

Refer to section 1.0.1 for internal backup battery temperature ranges.

## 2.10 Memory

| Parameter | Value |
|---|---|
| PSRAM | |
| ST 9100 | 8 MB |
| ST 9101 | 8 MB with Octo SPI interface |
| External Flash | |
| ST 9100 | 16 MB |
| ST 9101 | 32 MB |

## 2.11 Environmental

| Parameter | Description |
|---|---|
| Vibration | The terminal meets all its specifications during exposure to random vehicular vibration levels per SAE J1455, section 4.10.4.2 figures 6, 7, and 8, and MIL-STD-810H, section 514.8, figure 514.8C-1. |
| Mechanical Shock | The terminal meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 20 G and durations of 11 ms as specified in MIL-STD-810H, section 516.8, Procedure I, section 2.3.2c. |
| Thermal Shock | The terminal meets all of its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2 |
| Drop Test | The terminal meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1. |
| Altitude | The terminal meets all specifications after a nonoperating 12.2 km (7.5 miles) altitude test as detailed in SAE J1455, section 4.9.3, except with an ambient temperature of -40°C (-40°F). |
| Humidity | The terminal meets all its specifications during exposure to 90% relative humidity at +85°C (185°F), per the test methodology of SAE J1455, section 4.2.3 (3 x 8-hour humidity cycle per figure 4a) |
| Ingress Protection | IP67 – The terminal meets all of its specifications after immersion and dust tests as detailed in IEC 60529, sections 13.1, 13.4, 14.1, 14.2.7 and 14.3 (with and without optional terminal shroud) |
| ESD (Enclosure) | All electrical interfaces operate normally after being subjected to 8 kV ESD contact discharge per IEC 60945 and IEC 61000-4-2 human body model, level 3. |


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# 2.12 Sensors

## 2.12.1 Temperature Sensor

| Parameter | Value |
|---|---|
| Range | -40 to +85°C (-40 to +185°F) |
| Accuracy (typical) | ±4°C (±7.2°F) |

## 2.12.2 Accelerometer

The transceiver has a 3D 16-bit accelerometer to detect motion in any axis.

In low power applications, frequent GPS fixes can dominate the power budget. To reduce the power budget effects of GPS fixes, the accelerometer can be used to detect if motion has occurred.

The accelerometer thresholds to detect advanced features such as driver behavior monitoring vary depending on the environment. To avoid false detects, extensive testing is required to ensure that adequate acceleration magnitude thresholds and time durations are used.

Refer to T405 for additional details.

For ST 9100 :

| Parameter | Condition | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|
| Resolution | - | - | 16 | - | bit |
| Acceleration Range | software selectable | - | ±2 | - | g |
| | | - | ±4 | - | g |
| | | - | ±8 | - | g |
| | | - | ±16 | - | g |
| Sensitivity | 2 g | - | 16384 | - | LSB/g |
| | 4 g | - | 8192 | - | LSB/g |
| | 8 g | - | 4096 | - | LSB/g |
| | 16 g | - | 2048 | - | LSB/g |
| Sensitivity Temperature Drift | 3V3 supply | - | ±0.03 | - | %/K |
| Zero-g Offset Component level | Ta = 25°C | - | 25 | - | mg |
| Zero-g Offset Temperature Drift | 3 V supply | - | ±1.0 | - | mg/K |
| Wake up Time | from low power or suspended modes | - | 0.8 | - | ms |
| Start up Time | from low power or suspended modes to normal | - | 3.2 | 3.8 | ms |


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For ST 9101:

| Parameter | Condition | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|
| Resolution | - | - | 16 | - | bit |
| Acceleration Range | software selectable | - | ±2 | - | g |
| | | - | ±4 | - | g |
| | | - | ±8 | - | g |
| | | - | ±16 | - | g |
| Bandwidth Filtering | selectable via digital interface | 8 | - | 1000 | Hz |
| Sensitivity | 2 g | - | 16384 | - | LSB/g |
| | 4 g | - | 8192 | - | LSB/g |
| | 8 g | - | 4096 | - | LSB/g |
| | 16 g | - | 2048 | - | LSB/g |
| Sensitivity Temperature Drift | 3V3 supply | - | 0.004 | - | %/K |
| Zero-g Offset over Lifetime | Ta = 25°C | - | ±150 | - | mg |
| Zero-g Offset Temperature Drift | 3 V supply | - | ±0.25 | - | mg/K |
| Wake-up Time | from low power or suspended modes | - | 0.8 | - | ms |
| Start up Time | from low power or suspended modes to normal | - | 2 | - | ms |

## 2.13 Cellular Module

### 2.13.1 ST 9100 LTE Transceivers Operating in the Americas

Transceivers (p/n ST9100-D01), operating in North or South America, have the following characteristics.

| | |
|---|---|
| **Type** | u-blox Toby R200 series |
| **LTE Module** | LTE bands: 2, 4, 5, 12<br>UMTS bands: 850, 900, 1900, 2100 MHz<br>GSM 850/900/1800/1900 MHz |
| **Output Power** | LTE power Class 3 (23 dBm)<br>UMTS/HSDPA/HSUPA power Class 3 (24 dBm)<br>GSM/GPRS Power Class:<br>*Class 4 (33 dBm) for GSM/E-GSM band<br>*Class 1 (30 dBm) for DCSPCS band<br>EDGE(8-PSK) Power Class:<br>*Class E2 (27 dBm) for GSM/E-GSM band<br>*Class E2 (26 dBm) for DCS/PCS band |
| **Input Power** | Peak currents of 1.5 A typical, 1.9 A maximum. Module supply peak current consumption: peak of current consumption through the VCC pins during a GSM 1-slot Tx burst at maximum Tx power, with a matched antenna |


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| | |
|---|---|
| **Data Transfer** | LTE Category 1: up to 10.3Mb/s DL, 5.2 Mb/s UL<br>HSDPA category 8: up to 7.2 Mb/s DL, HSUPA category 6: up to 5.76 Mb/s UL<br>GPRS multi-slot class 33, CS1-CS4, up to 107 kb/s DL, up to 85.6 kb/s UL<br>EDGE multi-slot class 33, MCS1-MCS9, up to 296 kb/s DL, up to 236.8 kb/s UL |
| **Antenna Detect** | Output DC current pulse: 9 µA typical<br>Output DC current pulse time length: typical 330 µs |

### 2.13.2 ST 9100 LTE Transceivers Operating Outside of the Americas

Transceivers (p/n ST9100-C01), operating outside of North or South America, have the following characteristics.

| | |
|---|---|
| **Type** | u-blox Toby L280 series |
| **LTE Module** | LTE bands: 1, 3, 5, 7, 8, 28<br>UMTS bands: 850, 900, 1900, 2100 MHz<br>GSM 850/900/1800/1900 MHz |
| **Output Power** | LTE power Class 3 (23dBm)<br>UMTS/HSDPA/HSUPA power Class 3 (24dBm)<br>GSM/GPRS Power Class:<br>*Class 4 (33 dBm) for GSM/E-GSM band<br>*Class 1 (30 dBm) for DCSPCS band<br>EDGE(8-PSK) Power Class:<br>*Class E2 (27 dBm) for GSM/E-GSM band<br>*Class E2 (26 dBm) for DCS/PCS band |
| **Input Power** | Peak currents of 1.9 A typical, 2.5 A maximum. Module supply peak current consumption: peak of current consumption through the VCC pins during a GSM 1-slot Tx burst at maximum Tx power, with a matched antenna |
| **Data Transfer** | LTE Category 4: up to 150Mb/s DL, 50 Mb/s UL<br>HSDPA category 24: up to 42 Mb/s DL, HSUPA category 6: up to 5.6 Mb/s UL<br>GPRS multi-slot class 12, CS1-CS4, up to 85.6 kb/s DL/UL<br>EDGE multi-slot class 12, MCS1-MCS9, up to 236.8 kb/s DL/UL |
| **Antenna Detect** | Output DC current pulse: 21 µA typical<br>Output DC current pulse time length: typical 3.6 ms |

### 2.13.3 ST 9101 Transceivers

Transceivers (p/n ST9101-F01) have the following characteristics.

| | |
|---|---|
| **Type** | u-blox LARA R6001D |
| **LTE Module** | LTE FDD bands: 12, 28, 13, 20, 18, 19, 26, 5, 8, 4, 3, 2, 1, 7<br>LTE TDD bands: 39, 40, 41, 38<br>WCDMA bands: 5, 8, 2, 1<br>GSM bands: GSM 850, E-GSM 900, DCS 1800, PCS 1900 |


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| | |
|---|---|
| **Output Power** | LTE power Class 3 (23 dBm)<br>UMTS/HSDPA/HSUPA power Class 3 (24 dBm)<br>GSM/GPRS Power Class:<br>*Class 4 (33 dBm) for GSM 850/E-GSM 900 band<br>*Class 1 (30 dBm) for DCS 1800/PCS 1900 band<br>EDGE(8-PSK) Power Class:<br>*Class E2 (27 dBm) for GSM 850/E-GSM 900 band<br>*Class E2 (26 dBm) for DCS 1800/PCS 1900 band |
| **Input Power** | Peak currents of 1.5 A typical, 1.9 A maximum. Module supply peak current consumption: peak of current consumption through the VCC pins during a GSM 1-slot Tx burst at maximum Tx power, with a matched antenna |

## 2.14 LED

The transceiver has four visible LEDs (2.14) to indicate status. For more detailed LED information, refer to [T404].

Figure 17: LED Location




Table 5: LED Operation

| Icon | Function | Description | Color | LED ON | LED OFF |
|---|---|---|---|---|---|
| | Sensor | Indicates whether a central device (Bluetooth) is connected to the terminal or if the terminal is in fast advertising mode. | BLUE | Bluetooth Central Connected | Bluetooth Central Disconnected |
| | Cellular | Indicates cellular communications status.<br>Note: This LED only functions if the cellular module is powered on. The operation of this LED is dependent on the type of cellular module being used in the device. | GREEN | Data 2G/3G /Registered LTE | No Service (no network or not registered) |
| | Satellite / Satcom | Indicates satellite communications status. | YELLOW | -Power ON /External Reset | - |


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| Icon | Function | Description | Color | LED ON | LED OFF |
|---|---|---|---|---|---|
|  | Power | Indicates that the transceiver has external power. | RED | External Power Present | External Power Absent |

## 2.15 Mechanical

### 2.15.1 ST 91xx

| Parameter | Value |
|---|---|
| Weight | 465 g (16 oz) |
| Enclosure Material | Lexan plastic |

Figure 18: ST 91xx Top View Dimensions

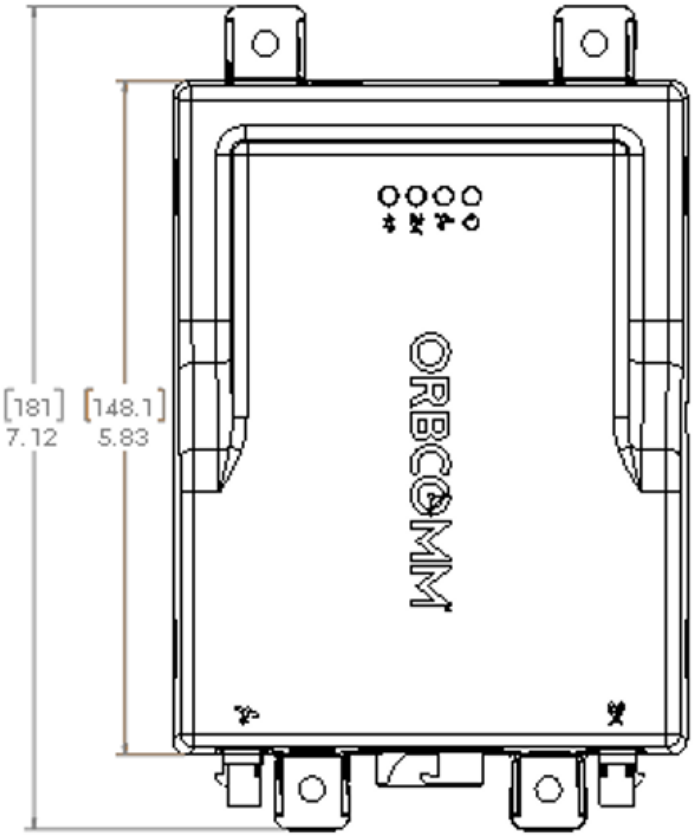
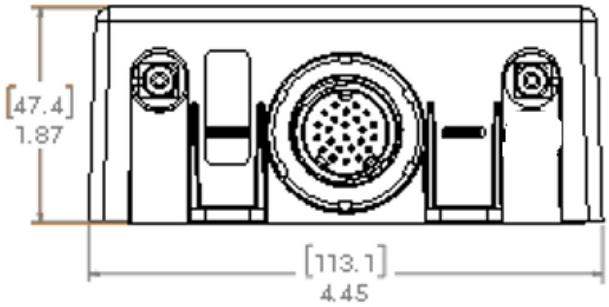


Figure 19: ST 91xx Side Connector View Dimensions



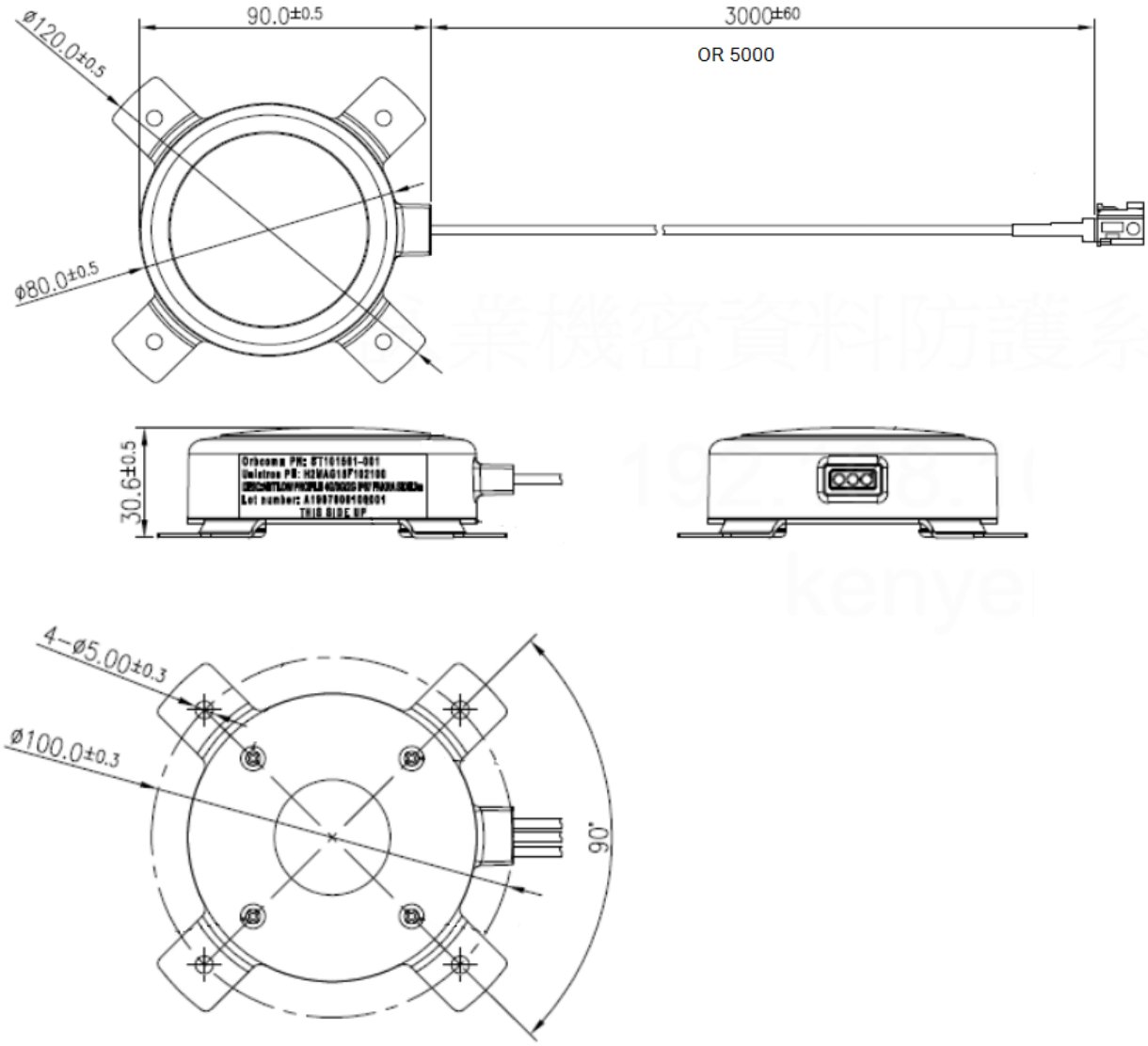


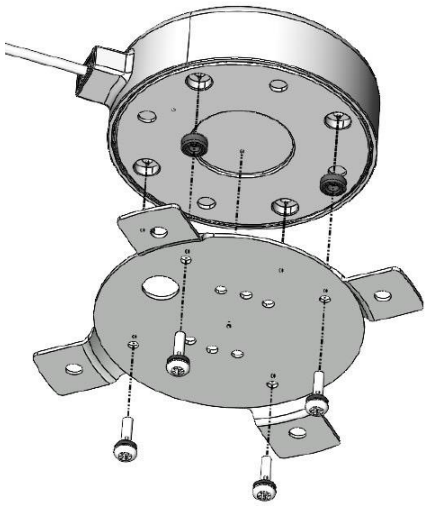
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## 2.15.2 Cellular Antennas

LTE/3G/2G Puck Cellular Antenna (ST101561-001 and ST101561-002)

Available with a 3 m cable (ST101561-001) or a 5 m cable (ST101561-002).

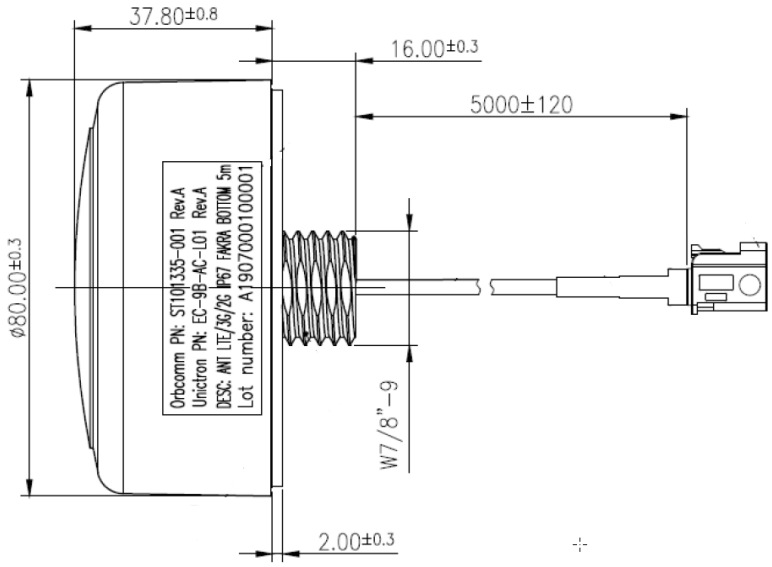




| Parameter | Value |
|---|---|
| Dimensions | Figure dimensions shown in mm |
| Cable Length | 3 m (10 ft.) and 5 m (16 ft.) (side exit) |
| Mounting | FAKRA straight plug connector |
| Operating / Storage Temperature | -40°C to 85°C (-40°F to 185°F) |
| Waterproof | IP67 |


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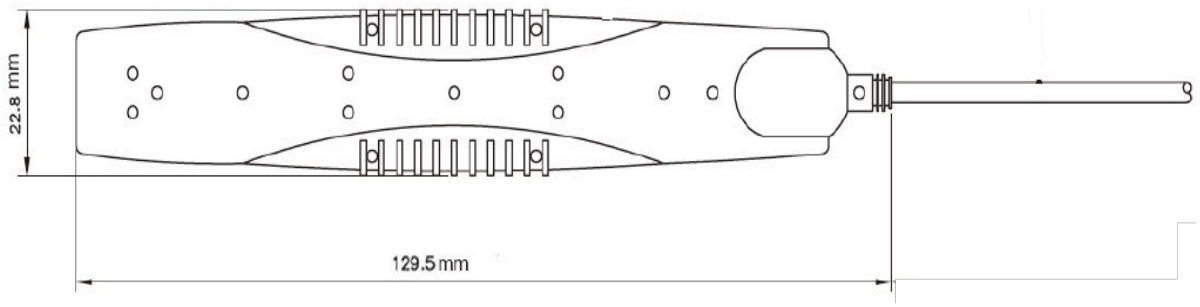
LTE Puck Cellular Antenna (ST101335-001)




| Parameter | Value |
|---|---|
| Dimensions | Figure dimensions shown in mm |
| Cable Length | 5 m (16 ft.) (bottom exit - screw mount) |
| Mounting | FAKRA straight plug connector |
| Operating / Storage Temperature | -40°C to 85°C (-40°F to 185°F) |
| Waterproof | IP67 |


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LTE/3G/2G Bar Cellular Antenna (ST101066-001)

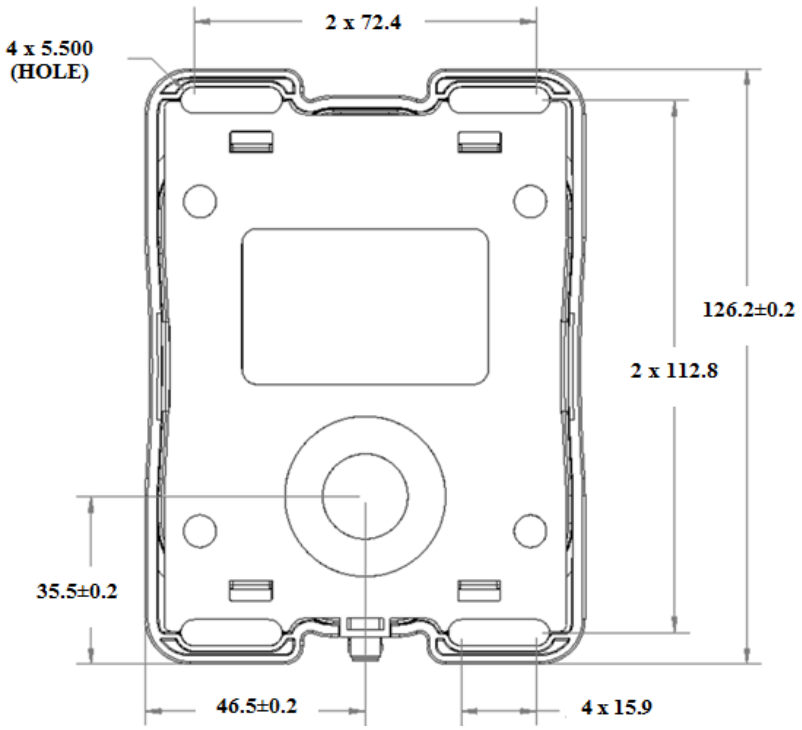



| Parameter | Value |
|---|---|
| Weight | 55 g (2 oz.) |
| Dimensions | 129.5 x 22.8 x 7 mm (5 in. x 0.9 x 0.27 in.) |
| Cable Length | 3 m (10 ft.) |
| Mounting | FAKRA straight plug connector |
| Operating Temperature | -40°C to 80°C (-40°F to 176°F) |
| Waterproof | IP65 |


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### 2.15.3 Satellite Antenna

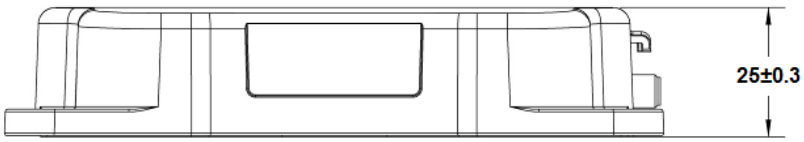
Figure 20: Satellite Antenna (standard and low elevation) - Bottom View (mm)




#### 2.15.3.1 Standard Antenna

| Parameter | Value |
|---|---|
| Weight | Side entry with 5 m (16 ft.) cable: 360 g (13 oz.) |
| Enclosure Material | Lexan EXL |
| Color Code | 8T9D076 (white) |
| Sealing Gasket Material | Santoprene® |

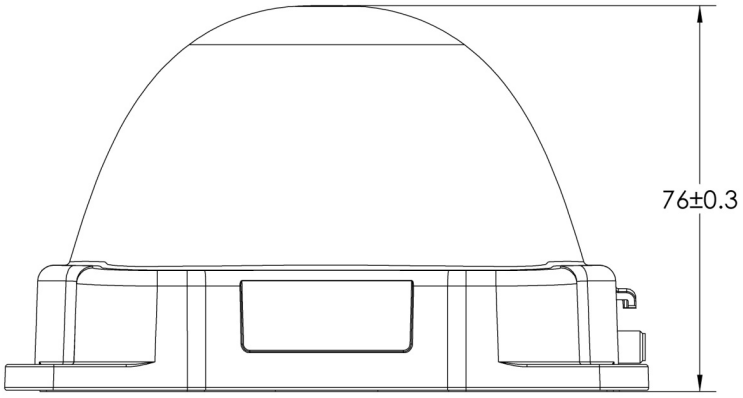
Figure 21: Standard Antenna Height Dimensions (mm)



### 2.15.3.2 Low Elevation Antenna

| Parameter | Value |
|---|---|
| Weight | Side entry with 5 m (16 ft.) cable: 365 g (13 oz.) |
| Enclosure Material | Lexan EXL |
| Color Code | 8T9D076 (white) |
| Sealing Gasket Material | Santoprene® |

Figure 22: Low Elevation Antenna Height Dimensions (mm)

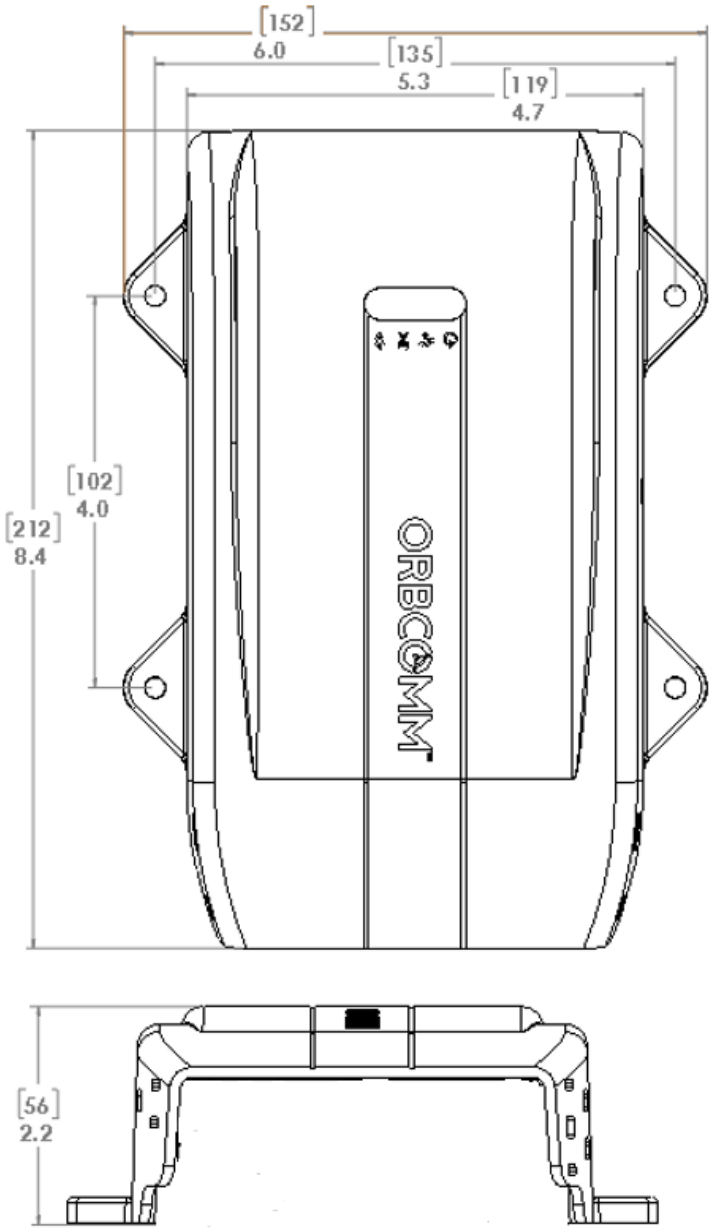



### 2.15.3.3 Terminal Shroud

| Parameter | Value |
|---|---|
| Weight | 150 g (2 oz.) |
| Enclosure Material | Lexan EXL |
| Color Code | 8T9D076 (white) |


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Figure 23: Terminal Shroud Dimensions



# 3 COMPLIANCE

Certifications for the following have been received, unless noted otherwise. Contact your Account Manger for updates.

## Inmarsat Type Approval

## Industry Canada

- ICES-003
- RSS-170, Issue 2, Spectrum Management and Telecommunications Policy, Radio Standard
- RSS-102, radiation safety per Safety Code 6 (compliance shown by computation)
- IC: 11881A-ST9100; 11881A-UNNB30; 8595A-TOBYL280 OR 8595A-1EHM44NN

## FCC

- CFR 47 Part 25, CFR 47 Part 15
- CONTAINS FCC ID: XGS-ST9100; XGS-UNNB30; XPYTOBYL280 OR XPY1EHM44NN

## CE RED 2014/53/EU

## Ingress Protection

- Cellular antenna: bar antenna: IP65; puck antenna: IP67
- Satellite antennas: IP67
- Transceiver unit: IP67

## RoHS

- Restriction of Hazardous Substances (RoHS) [1]

## IEC 60945

- International Standard that certifies navigation and radio communication equipment and systems for use in maritime application.

  Note: You must use the IEC 60945 compliant cable. Contact your Account Manager for part number details.

## Anatel Homologation

## UN

- UN 38.3 Transportation Compliance

[1] European Union's (EU) Directive 2002/95/EEC "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment.


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## PTCRB

## ITF

## ICASA

## AMAC

## EU Declaration of Conformity

Hereby, ORBCOMM Inc. declares that the radio equipment types listed in this document comply with Directive 2014/53/EU.

The full text of the EU declaration of conformity is available from http://www2.orbcomm.com/eudoc.

## WARNING:

- The minimum 20 cm (8 in.) separation distance from the device is required for RF exposure safety for all persons.
- Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.
- This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:(1) This device may not cause interference.(2) This device must accept any interference, including interference that may cause undesired operation of the device.
    - L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

      1) L'appareil ne doit pas produire de brouillage;

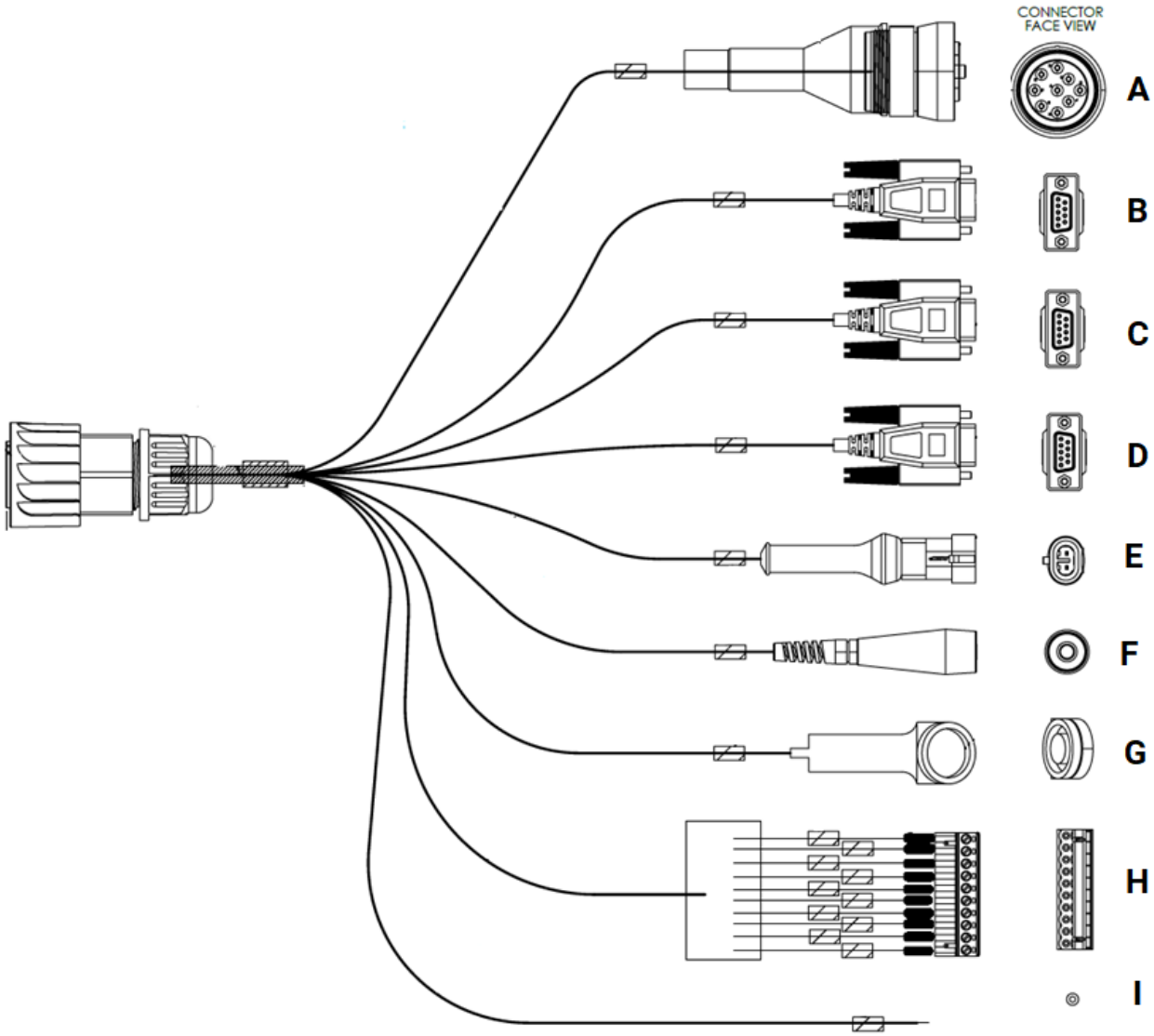
      2) L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.


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# APPENDIX A DEVELOPMENT CABLE

The development cable is p/n ST101084-001.

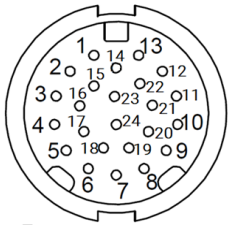
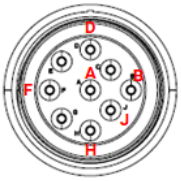
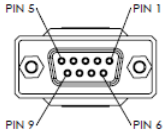
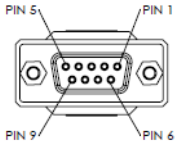
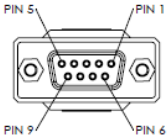
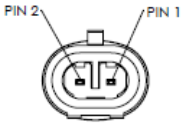
Figure 24: Development Cable



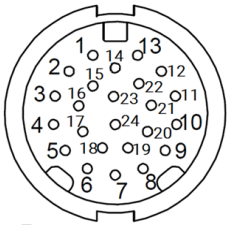
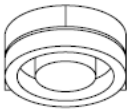
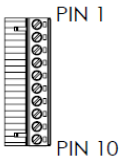
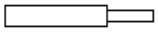


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Table 6: Development Cable Connectors

| Transceiver Connector | End A | End B | End C | End D | End E |
|---|---|---|---|---|---|
|  |  |  |  |  |  |
| PIN 9 - 1Wire Com | | | | | |
| PIN 10 - RS232 TX | | PIN 2 | | | |
| PIN 21 - RS232 RX | | PIN 3 | | | |
| PIN 22 - AUX RS232 TX | | | PIN 2 | | |
| PIN 13 - CAN 2 Low | PIN D | | | | |
| PIN 14 - CAN 1 Low | PIN J | | | | |
| PIN 1 - RS485 A | | | | PIN 1 | |
| PIN 24 - Dig IN 2 | | | | | |
| PIN 16 - Dig IN 1 | | | | | |
| PIN 4 - I/O_4 | | | | | |
| PIN 5 - I/O_2 | | | | | |
| PIN 6 _ Ground | PIN A and PIN E | PIN 5 | PIN 5 | PIN 5 | PIN 2 |
| PIN 7 - VEXT | PIN B | | | | PIN 1 |
| PIN 8 - Out 6 | | | | | |
| PIN 18 - I/O_1 | | | | | |
| PIN 17 - I/O_3 | | | | | |
| PIN 3 - Dig IN 3 | | | | | |
| PIN 15 - RS485 B | | | | PIN 2 | |
| PIN 23 - CAN 1 High | PIN H | | | | |
| PIN 12 - CAN 2 High | PIN C | | | | |
| PIN 11 - AUX RS232 RX | | | PIN 3 | | |
| PIN 20 - 1Wire Data | | | | | |
| PIN 19 - Out 5 | | | | | |
| PIN 2 - Dig IN 4 | | | | | |


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| Transceiver Connector | End F | End G | End H | End I |
|---|---|---|---|---|
|  |  |  |  |  |
| PIN 9 - 1Wire Com | | 1Wire Common | | |
| PIN 10 - RS232 TX | | | | |
| PIN 21 - RS232 RX | | | | |
| PIN 22 - AUX RS232 TX | | | | |
| PIN 13 - CAN 2 Low | | | | |
| PIN 14 - CAN 1 Low | | | | |
| PIN 1 - RS485 A | | | | |
| PIN 24 - Dig IN 2 | | | PIN 8 | |
| PIN 16 - Dig IN 1 | | | PIN 7 | |
| PIN 4 - I/O_4 | | | PIN 4 | |
| PIN 5 - I/O_2 | Input 1 | | | |
| PIN 6 _ Ground | Ground | | PIN 2 | |
| PIN 7 - VEXT | | | PIN 1 | |
| PIN 8 - Out 6 | | | PIN 6 | |
| PIN 18 - I/O_1 | | | | I/O_1 |
| PIN 17 - I/O_3 | | | PIN 3 | |
| PIN 3 - Dig IN 3 | | | PIN 9 | |
| PIN 15 - RS485 B | | | | |
| PIN 23 - CAN 1 High | | | | |
| PIN 12 - CAN 2 High | | | | |
| PIN 11 - AUX RS232 RX | | | | |
| PIN 20 - 1Wire Data | | 1Wire Data | | |
| PIN 19 - Out 5 | | | PIN 5 | |
| PIN 2 - Dig IN 4 | | | PIN 10 | |


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